

HOMEWORK 7

Track 2

Due date: November 26, 2023.

Reading assignment. Revise chapters 1, 2 and 3 of the class lecture notes. Read chapter 5 of the class lecture notes. Read chapters 6 & 7 of the text book. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

Problem 1.

- a) Show that a real 2×2 matrix A is a solution of the equation $A^2 = -I$, if and only if,

$$A = \begin{pmatrix} \pm\sqrt{pq-1} & -p \\ q & \mp\sqrt{pq-1} \end{pmatrix},$$

with $pq \geq 1$, and the signs in the double signs should be chosen simultaneously as the top or bottom ones.

- b) Show that there is no real 2×2 matrix A that satisfies the equation

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 - \varepsilon \end{pmatrix},$$

when $\varepsilon > 0$.

Problem 2. Find a real matrix B such that

$$B^4 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

Problem 3. Find all integer values of n for which there exists a complex matrix B such that

$$B^n = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Problem 4. We say that a matrix is **Hermitian** if $A^H = A$. Show that a Hermitian matrix has only real eigenvalues. *Hint:* Look at $v^H A v$ with v being an eigenvector.

Problem 5. Show that the only Hermitian matrix A that satisfies the equation

$$A^5 + A^3 + A = 3I$$

is $A = I$. *Hint:* The function x^{2n+1} is monotonic increasing for all real values of x and n a non-negative integer.

Problem 6. Find the Jordan decomposition of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$